

Pranavi Kondapalli

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EDUCATION

Carnegie Mellon University School of Computer Science

Pittsburgh, PA

Bachelor of Science in Artificial Intelligence, GPA: 3.4

May 2027

Courses: Machine Learning, Deep Learning, Design and Analysis of Algorithms, Parallel and Sequential Data Structures and Algorithms, Probability and Computing, Ethics and Policy Issues in Computing, Computer Systems

EXPERIENCE

Customer Success Engineer Intern

New York, NY

IBM

August 2026 – Present

Machine Learning Researcher

Pittsburgh, PA

Machine Learning for Speech Processing Lab, CMU

May 2026 – August 2026

- Develop a speaker class augmentation framework in PyTorch that generates artificial speaker embeddings through synthetic mixup, increasing dataset diversity for speaker verification models.
- Design a training pipeline to align real and synthetic speaker embeddings, reducing speaker verification EER from 3.84% to 3.52% on VoxCeleb1 evaluation.

AI Software Engineer Intern

Remote

AI-CARING Institute

March 2026 – August 2026

- Create a socially aware conversational AI system to infer latent user values from personal narratives and enable context aware responses.
- Implement a hierarchical RAG pipeline that converts dialogue into a knowledge graph and retrieves long-term user context for personalized response generation.
- Benchmark LLM performance on 2,000+ narratives, validating narrative based value inference as a scalable approach for user-centered AI systems.

AI Researcher – Computer Vision and Robotics

Pittsburgh, PA

Reliable Autonomous System Lab, CMU

May 2024 – March 2026

- Built a real-time RGB-D perception system in Python and OpenCV for multiple object detection, 3D localization, and tracking in dynamic environments.
- Improved object tracking system's accuracy by 21% by adding fault tolerant logic to preserve object identity across occlusions, using pose tracking, cabinet open/close detections, and 3D consolidation.
- Evaluated system performance against a SOTA multiple object tracking baseline, achieving 66.9% precision vs 18.4% baseline and 98.3% accuracy vs 32.0% baseline.
- Co-first author on paper under review at IEEE/CVF WACV 2027

PROJECTS

OctoMind – AI Task Scheduler | *Next.js, Gemini API, TypeScript, Prisma, PostgreSQL*

- Developed an AI-powered task manager that predicts task duration from language input and automatically schedules work.
- Implemented backend workflows for task duration prediction, deadline-aware prioritization, and automatic calendar scheduling based on user constraints and feedback.
- Designed database schemas and API logic to support user authentication and dynamic schedule updates.

Think Before You Trust – LLM Security Detection Framework | *Python, AI Security*

- Developed an adversarial evaluation pipeline analyzing prompt injection attacks and Chain-of-Thought reasoning behavior across LLM systems, leveraging 1,400+ generated prompt traces.
- Built binary classifiers on CoT embeddings using SVM, Logistic Regression, and Random Forest models, achieving up to 0.974 AUC in detecting injected vs. non-injected prompts.
- Evaluated prompt injection robustness under CoT and non-CoT conditions, demonstrating that CoT reasoning significantly reduced attack success rates across multiple prompt leaking attacks.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript/TypeScript, Java, SQL, R

Frameworks/Libraries: PyTorch, TensorFlow, OpenCV, Node.js, Next.js, React.js

AI Tools: HuggingFace Transformers, FAISS, BM25, Sentence-Transformers, Grounding DINO, SAM

Cloud/DevOps: Docker, Git, REST APIs, AWS